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Chapter 1: Computers and Intelligence

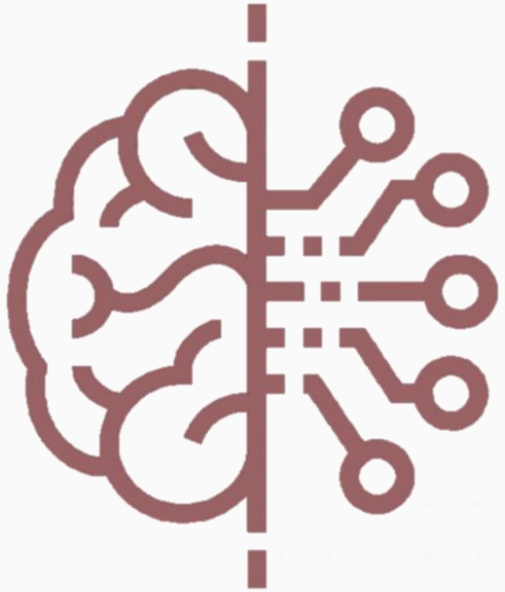
SCSE3103: Cognitive Computing
Faculty of Computing (FC), UTM

Innovating Solutions



UTM JOHOR BAHRU

Content



- Thinking and intelligence
- AI foundations and types
- Responsible AI and governance
- Computational intelligence
- AI applications

When does a computer seem intelligent?

Choose one system you used this week

- Navigation app
- Search or recommendation
- Chatbot or writing assistant
- Face or voice recognition

Discuss with a partner

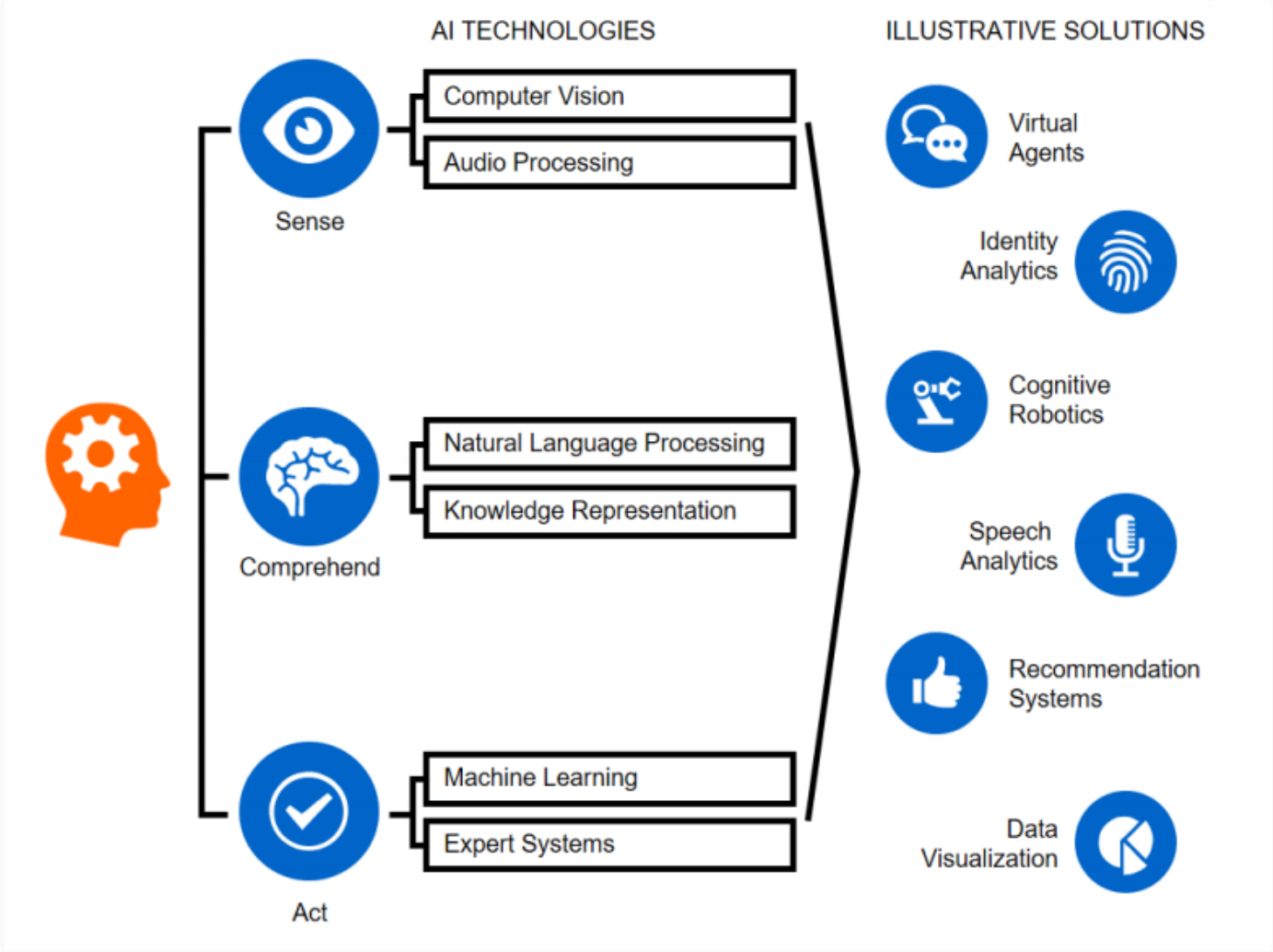
- What input did it receive?
- What output did it produce?
- What made the behaviour look intelligent?
- Where could it fail?

What comprises to Artificial Intelligence?

Artificial Intelligence is not just a part of computer science even it's so vast and requires lots of other factors which can contribute to it. To create the AI first we should know that how intelligence is composed, so the Intelligence is an intangible part of our brain which is **a combination of reasoning, learning, problem-solving perception, language understanding, etc.**



Area of AI?


AI TECHNOLOGIES
ILLUSTRATIVE SOLUTIONS
Computer Vision
Audio Processing
Sense
Natural Language Processing
Knowledge Representation
Comprehend
Machine Learning
Expert Systems
Act
Virtual Agents
Identity Analytics
Cognitive Robotics
Speech Analytics
Recommendation Systems
Data Visualization

Thinking as information processing

01 Perception

Acquire signals from the environment, such as text, images, sound, or sensor data.

02 Representation

Organise information into concepts, symbols, features, or internal states.

03 Reasoning

Combine evidence and knowledge to infer, compare, predict, or plan.

04 Action and learning

Choose an action, observe its result, and update future behaviour.

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Forms of human thinking

Fast and intuitive

Pattern recognition, habits, and rapid judgements. Efficient but sensitive to bias.

Deliberate and analytical

Step-by-step reasoning, calculation, comparison, and explicit checking.

Creative and reflective

Generating alternatives, reframing problems, and examining one's own reasoning.

Computers can support each form, but performance on one task does not prove general understanding.

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Activity: Evidence of intelligence

Rank the behaviours from weakest to strongest evidence of intelligence

A A calculator returns the correct answer

B A robot adapts its path around an obstacle

C A model explains a medical image correctly

D A chatbot asks a useful clarifying question

Defend your ranking. Which criterion matters most: correctness, adaptation, explanation, or social interaction?

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Definitions of intelligence

Psychological view

Capacity to learn, reason, solve problems, adapt to new situations, and use knowledge effectively.

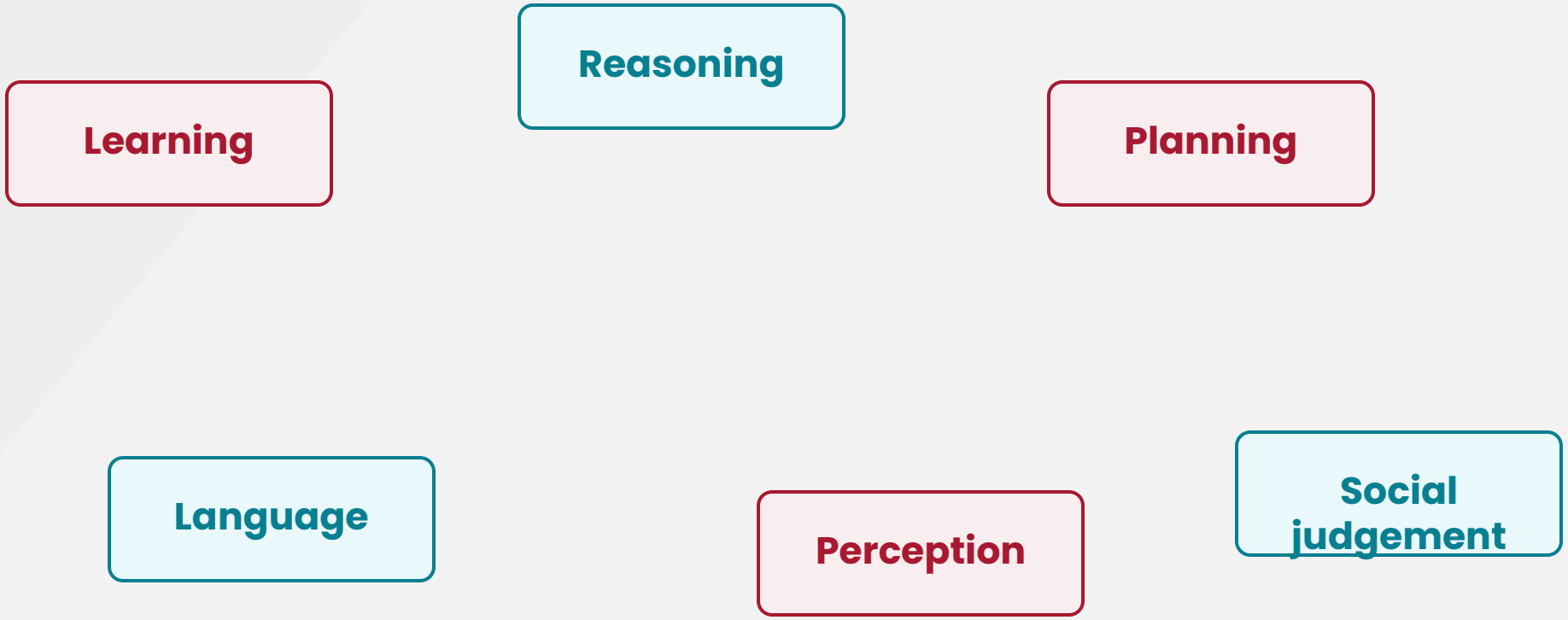
Agent view

Ability to achieve goals across a range of environments by choosing actions from available information.

A useful definition states the capability, the context, and the standard of success.

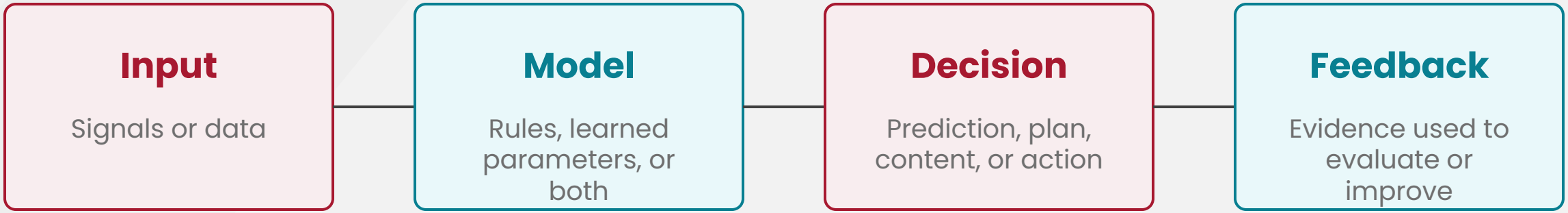
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Intelligence has several dimensions



An AI system can be strong in one dimension and weak in another.

From computation to intelligent behaviour



Intelligence claims require evidence across the whole loop, especially under new conditions.

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Turing's imitation game

“Can machines think?”

Turing replaced a difficult philosophical question with an observable test of conversational behaviour.

What the test probes

- Language use
- Context handling
- Plausible reasoning
- Human judgement of behaviour

Passing a behavioural test does not settle questions about consciousness or understanding.

The Dartmouth proposal and the birth of AI

1955 proposal

McCarthy, Minsky, Rochester, and Shannon proposed a 1956 summer study on machine intelligence.

Core ambition

Describe learning and intelligence precisely enough that a machine could simulate them.

The proposal named a research field and set an ambitious engineering agenda.

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A current working definition of AI

An AI system receives inputs and infers how to generate outputs such as predictions, content, recommendations, or decisions.

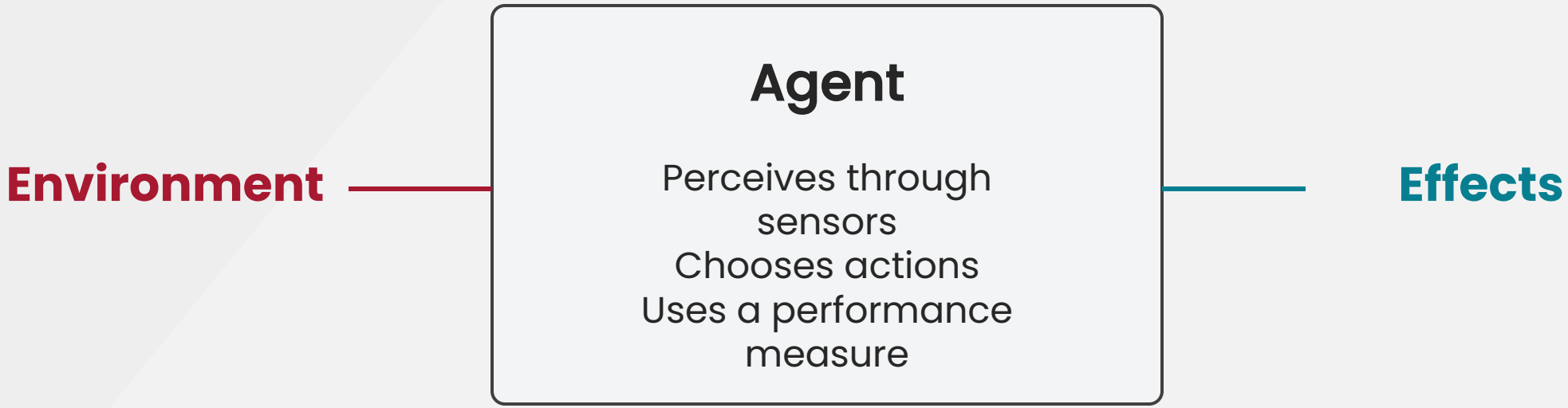
The outputs can influence physical or virtual environments. Systems differ in autonomy and in how much they adapt after deployment.

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What is Cognitive Computing?

Cognitive Computing refers to computer systems designed to **simulate aspects of human cognitive processes**, such as learning, reasoning, understanding language, recognizing patterns, interpreting context, and supporting decisions.

The rational-agent view



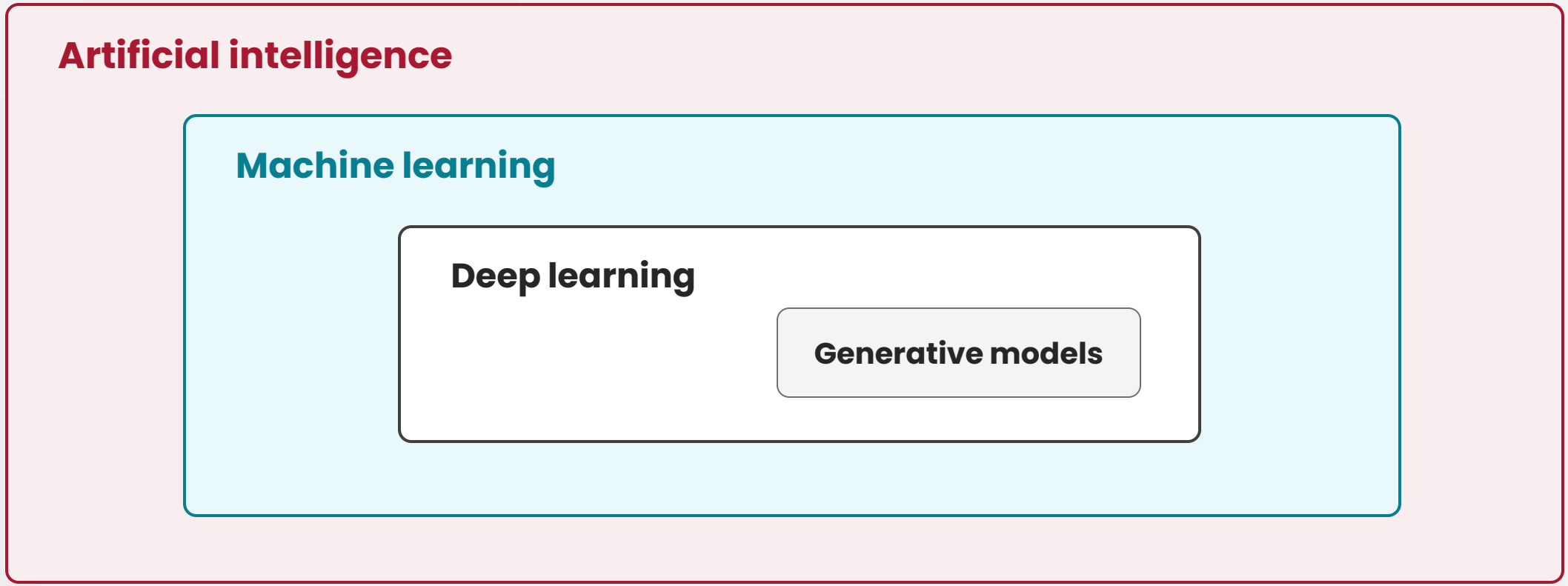
Rational means choosing an action expected to achieve the goal, given available information.

Capability scope: Narrow AI, AGI, and Super intelligence

Narrow AI	Performs bounded tasks within a defined domain	**Deployed today**
Artificial general intelligence(AGI)	Would transfer learning and reasoning across many domains	**Research goal, no accepted demonstration**
Super intelligence	Hypothetical capability exceeding humans across most cognitive tasks	**Speculative concept**

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AI, machine learning, deep learning, and generative AI



The terms overlap, but they describe different levels of scope, methods, and outputs.

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AI, machine learning, deep learning, and generative AI

Artificial Intelligence (AI) —computer systems that can perform tasks that normally require human intelligence, such as reasoning, problem-solving, planning, understanding language, and decision-making. AI includes both learning-based and rule-based approaches.

Machine Learning (ML) —computers **learn patterns from data** rather than being explicitly programmed with every rule. Examples include spam detection, recommendation systems, and predicting house prices.

Deep Learning (DL) —uses **multi-layered artificial neural networks** to learn complex patterns, typically from large amounts of data. It is widely used in image recognition, speech recognition, natural language processing, and autonomous systems

Generative Models — Models that learn patterns or distributions in training data and can **generate new content or samples** resembling that data. They can generate text, images, audio, video, or other data. Modern examples include large language models (LLMs) and diffusion models.

AI, machine learning, deep learning, and generative AI

However, **not every generative model is necessarily a deep-learning model**. Classical approaches such as Gaussian mixture models and Hidden Markov Models can also be generative.

Symbolic and data-driven AI

Symbolic approaches

- Explicit rules and representations
- Search, logic, planning, expert systems
- Often easier to inspect
- Can be brittle when rules or knowledge are incomplete

Data-driven approaches

- Patterns learned from examples
- Statistical models, neural networks, reinforcement learning
- Handles complex perception well
- Depends on data quality and evaluation

Hybrid systems combine learned perception with rules, search, or human constraints.

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Core capabilities in an AI system

Represent	Search and plan	Learn	Communicate
Encode states, objects, relations, and goals	Explore possible actions and consequences	Estimate patterns or policies from data and feedback	Interpret and generate language or other media

System design asks which capabilities are needed and how their errors interact.

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Checkpoint: Classify the system

A university helpdesk chatbot retrieves policy documents, ranks passages, and generates a reply.

Identify

- Inputs and outputs
- Learned components
- Rule-based components

Evaluate

- Success measure
- Likely failure
- Needed human oversight

Share one answer before the break

Break

10 minutes

When you return, bring one example of an AI system that could cause harm if it fails.

Responsible AI starts with purpose and impact

Responsible AI

A disciplined approach to designing, deploying, and governing AI so that benefits are pursued while risks to people, organisations, and society are managed.

Three questions

- Should this problem use AI?
- Who benefits and who may be harmed?
- How will we detect, respond to, and learn from failures?

Sources of AI risk

Data	Poor coverage, historical bias, privacy loss, or weak provenance
Model	Error, opacity, instability, security vulnerabilities, or misuse
Human use	Automation bias, inadequate skills, overreliance, or malicious use
System context	Wrong objective, unequal access, feedback loops, or weak governance

Risk depends on both technical performance and the setting in which people use the system.

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Case: Fairness in automated scholarship screening

Scenario

A model ranks applicants using past award decisions. Historical access to enrichment activities differs across regions and income groups.

Group analysis

- What outcome should fairness address?
- Which groups need separate evaluation?
- What evidence would show unequal impact?
- What human review or appeal is needed?

Fairness is a context-dependent choice that must be documented and tested.

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Common principles for trustworthy AI

Human benefit and rights

Purpose, dignity, inclusion, and proportionality

Fairness

Identify and reduce unjustified disparate outcomes

Transparency

Communicate AI use, limits, and relevant explanations

Safety and robustness

Test normal use, foreseeable misuse, and failure modes

Privacy and security

Protect data and systems throughout the lifecycle

Accountability

Assign responsibility, documentation, oversight, and redress

Malaysia's seven AIGE principles

1. **Fairness**
2. **Reliability, safety and control**
3. **Privacy and security**
4. **Inclusiveness**
5. **Transparency**
6. **Accountability**
7. **Human benefit and happiness**

AIGE addresses developers, deployers, and end users across the AI lifecycle.

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Responsible AI across the lifecycle

1 Plan and design

Define purpose, stakeholders, and acceptable risk

2 Collect and process data

Check consent, provenance, quality, and representation

3 Build and use the model

Document choices and apply constraints

4 Verify and validate

Test performance, fairness, safety, and misuse

5 Deploy

Set human roles, instructions, and fallback

6 Operate and monitor

Monitor drift, incidents, complaints, and retirement

Activity: A lightweight AI impact assessment

Scenario: An AI tool predicts which students may need academic support.

Purpose	What decision will the prediction support?	**People**	Who is affected, including people missing from the data?
Risk	What harms could follow from error, misuse, or disclosure?	**Evidence**	Which tests and subgroup results are required?
Control	Who reviews, overrides, explains, and handles appeals?	**Monitoring**	Which signals would trigger investigation or retirement?

Teams choose one risk and propose one preventive control plus one monitoring signal.

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Transparency, explanation, and human oversight

Transparency

Disclose when AI is used, its purpose, data boundaries, and known limitations.

Explanation

Provide information appropriate to the audience and the consequence of the decision.

Oversight

Give people authority, competence, time, and a safe way to intervene or appeal.

Human oversight fails when the reviewer lacks information, authority, or time.

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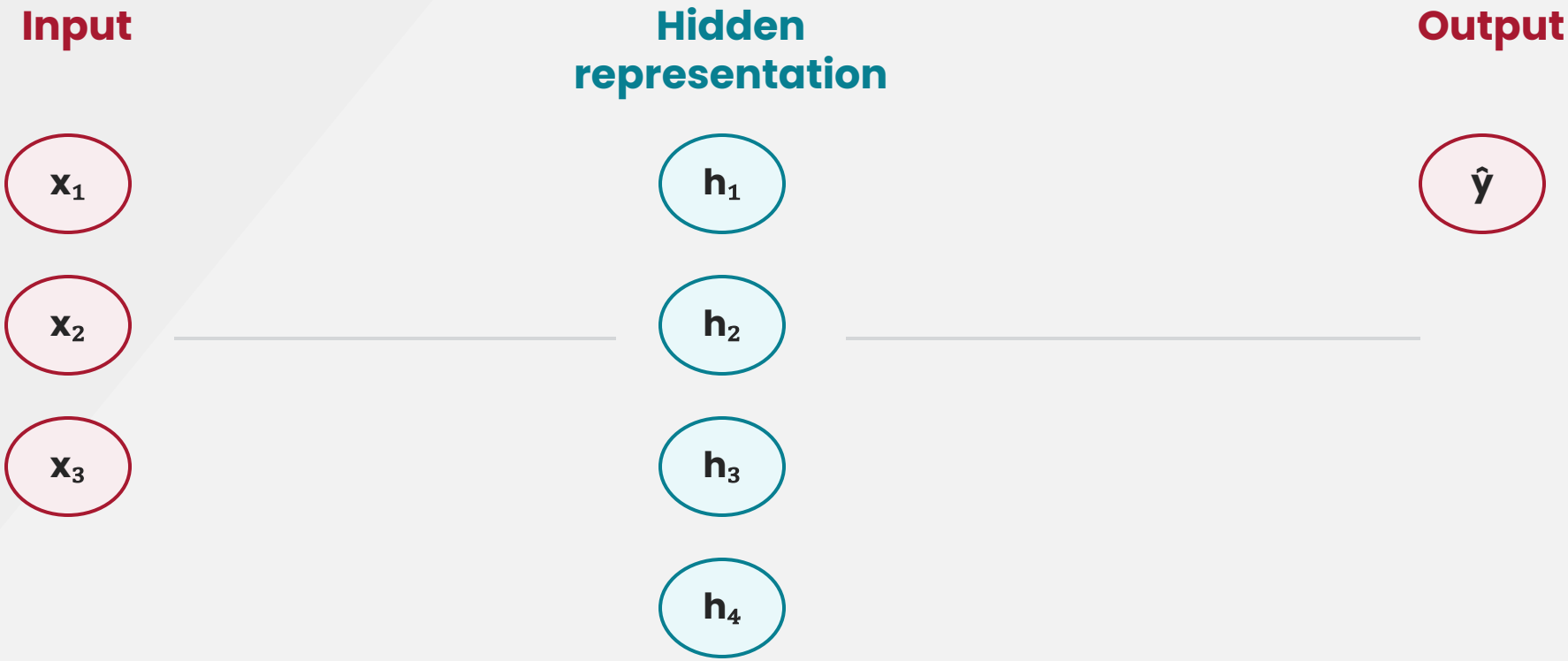
Computational Intelligence

Computational intelligence develops adaptive methods that learn from data, tolerate uncertainty, and search complex solution spaces.

- Neural computation learns representations and mappings
- Fuzzy systems reason with graded concepts
- Evolutionary computation searches by variation and selection

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Neural networks learn useful mappings



Training adjusts parameters to reduce error on examples. Evaluation checks whether the mapping generalises.

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Fuzzy logic represents degrees of membership

Crisp rule

Temperature ≥ 30°C means HOT

Membership changes abruptly at the threshold.

Fuzzy rule

A temperature can be partly WARM and partly HOT

Controllers combine graded rules to produce smooth actions.

Fuzzy logic models vagueness in concepts. Probability models uncertainty about events.

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Evolutionary computation searches by variation and selection

1 Population

Start with many candidate solutions

2 Evaluate

Score each candidate using a fitness function

3 Vary

Create new candidates through mutation and recombination

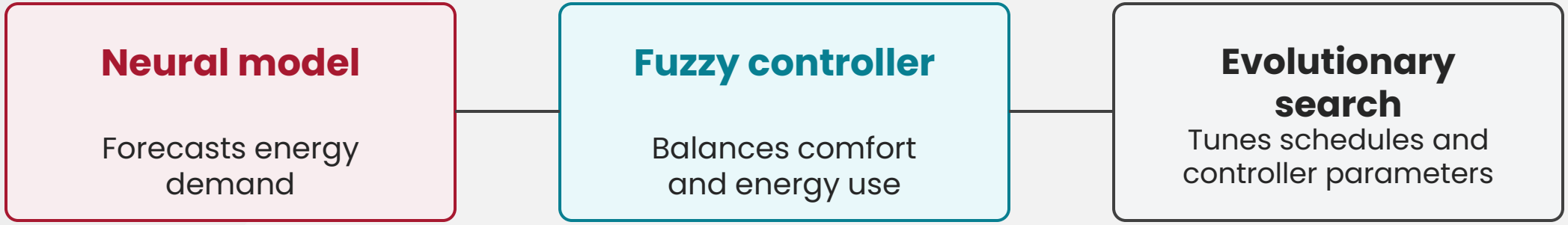
4 Select

Retain stronger candidates and repeat

Useful when the search space is large, irregular, or difficult to optimise with gradients.

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Hybrid computational intelligence



Example: a smart-building system combines prediction, interpretable control rules, and optimisation.

AI applications are patterns of input, inference, and action

Perception	Images, audio, sensors	Detect, classify, estimate
Prediction	Records and time series	Forecast risk, demand, or failure
Recommendation	Preferences and context	Rank options or personalise
Generation	Prompts and reference data	Produce text, code, images, or designs
Autonomous action	State and feedback	Plan and act in an environment

The same model may support different applications, with different consequences and controls.

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Case study: AI in science and health

AlphaFold

Deep learning predicts protein structures from amino-acid sequences, supporting structural biology and hypothesis generation.

Responsible use

- Communicate confidence and limitations
- Validate for the intended scientific or clinical use
- Protect sensitive health data
- Keep accountable human judgement

High predictive accuracy does not automatically establish clinical safety or usefulness.

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Applications across sectors

Education	Feedback, tutoring support, accessibility, administration	**Business**	Demand forecasting, customer support, quality inspection
Public services	Document processing, service routing, anomaly detection	**Transport and cities**	Traffic prediction, route planning, asset monitoring
Agriculture	Crop monitoring, disease detection, precision inputs	**Cybersecurity**	Threat detection, triage, and response support

Value comes from fitting the system to a real workflow, not from adding AI to every task.

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Chapter summary

- Intelligence involves several capabilities and must be judged in context.
- AI systems infer outputs that influence physical or virtual environments.
- Responsible AI connects principles to controls throughout the lifecycle.
- Computational Intelligence is one family of approaches used to achieve AI, especially through learning, adaptation, uncertainty handling, and nature-inspired computation.
- A valuable application links a real need to evidence, human roles, and monitoring.

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Self Review

1. Why can a system be intelligent at one task but not generally intelligent?

2. Name one difference between symbolic AI and machine learning.

3. Which AIGE principle is most relevant to an appeal process, and why?

4. How do fuzzy logic and probability represent different ideas?

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References and further reading

Foundations

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